Project D485 Revision

Task A

The company is currently facing a least-privileged issue because team members across departments can access each other's files and systems, posing a security concern. Also, the company needs the ability to back up data and enforce a retention period. U.S. federal government Regulatory and standard requirements are expected to be met. The team is expected to use appropriate resource tagging to identify and manage team resource costs. Encryption at rest and in transit must be implemented for all resources as part of the project requirements.

Task B

Identification of the service model: The proposed service model needs is Infrastructure as a Service (IAAS). Because, based on business needs, the company needs to implement its own custom security hardening, based on NIST 800-53 (which the federal government mostly uses as a baseline) and PCI DSS. To enforce these requirements and others, IAAS is the best service model to be considered in this scenario.

applicable regulatory compliance directives: Abiding by Federal law, such as FISTMA, is critical to business. And since the company maintains contracts with the U.S. government and processes card transactions daily, the Federal Information Security Modernization Act (FISMA) and the Payment Card Industry Data Security Standard (PCI DSS) requirements must be built into the architecture of the company's system.

Security Benefits and Challenges of Transitioning to this Service Model: While this is the best service model for our use case, it has its own security benefits and challenges. This option provides us the ability to customize further the Infrastructure for security hardening, underlying resource customization, and microservices integration management. For instance, we can funnel the logs from a particular server for constant and further analysis of events. For security hardening, we can configure encryption and related controls for data at rest and in transit, as specified in NIST 800-53 and PCI DSS guidelines and requirements, to satisfy regulatory requirements.

Challenges of Transition to this Service Model: While this cloud service model offers significant benefits for our business use case at SWBTL LLC. It is also important to address the transition challenges to cloud using this model. One challenge that could be faced is data migration complexity. If data migration is not properly managed, it could lead to data loss, corruption, and disruptions to daily operations. The database type that SWBTL LLC uses on premises must be compatible with the one that will be transitioned to in the cloud. And because

we are using IAAS, SWBTL LLC would be responsible for the heavy lifting of this process under the cloud provider's shared responsibility model (Azure).

Another Challenge with this model is application compatibility. The on-premise systems may not have the same system specifications as the cloud-based systems, and this could lead to the need for refactoring of the applications, because the packages, libraries, and utilities needed for the application may not be present by default with the cloud-based system, and there might be a need for refactoring of the application.

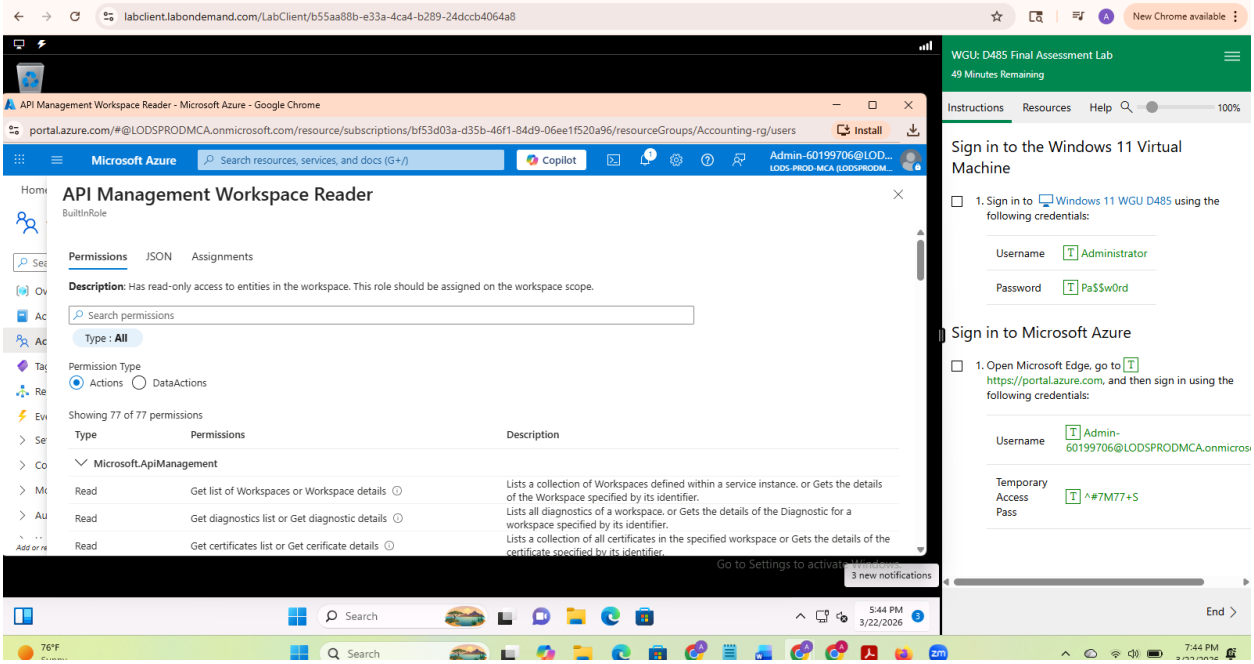
Lastly, another challenge with this cloud service model is the cloud skill gap among employees. If the company's IT team lacks adequate Cloud skills to manage security architecture and operations in the cloud, this could hinder the achievement of a well-architected cloud framework. Also, this can lead to an inappropriate security posture for the company's cloud architecture and deployment environments.

Task C

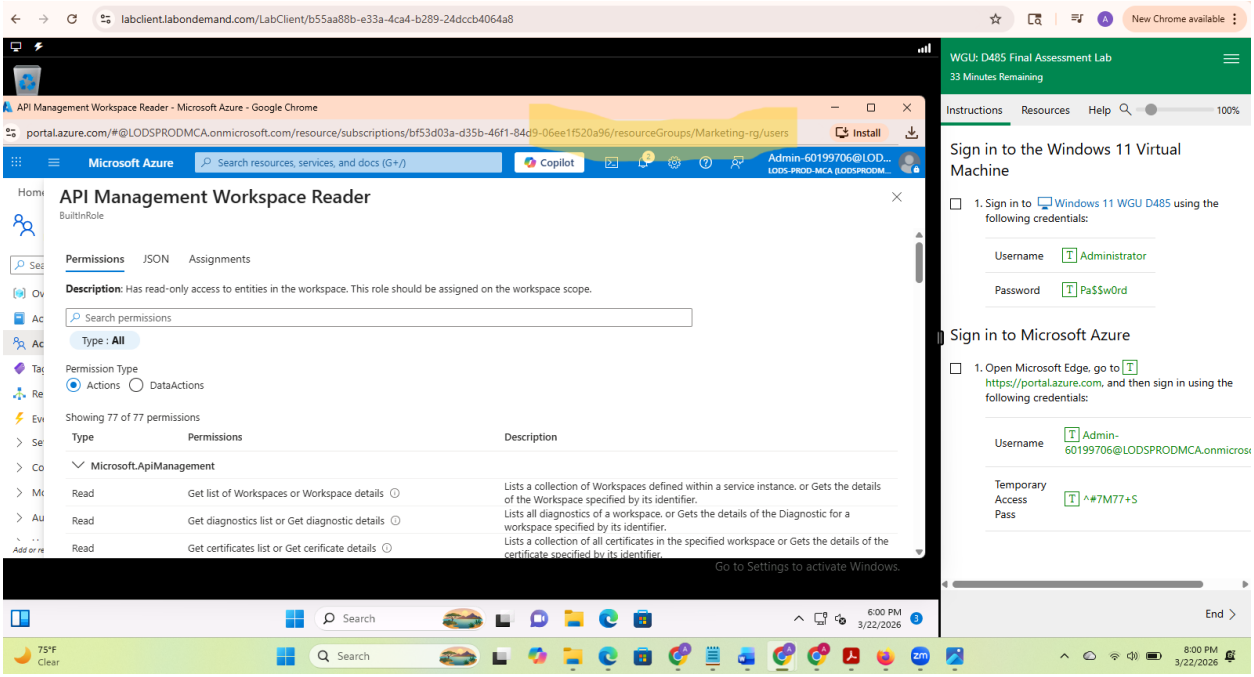
Given the current state of the environment, I recommend granting read-only access to the Marketing-rg and Accounting-rg workspaces. The reason for these two changes is to implement the principle of least privilege. This way, both the Accounting and marketing teams would have access only to their workspaces, with read-only access since they do not need to provision or manage resources. With this approach, all the members of the same department will have the same permission. However, any future exceptional role can be evaluated individually for additional permission if needed (For instance, a director, assistant director, etc., can have additional policy added (inline policy) based on their role’s needs). This implementation would help enforce the least-privileged rule and enhance accessibility within the department's workspace.

IT-rg was granted contributor-level privileged permissions. This is because this group would be responsible for resource provisioning, infrastructure customization, configurations, and overall IT infrastructure management. They need this permission within their workspace to fulfill their roles. And this approach would help enforce least privilege and ensure we are not relying on security by obscurity.

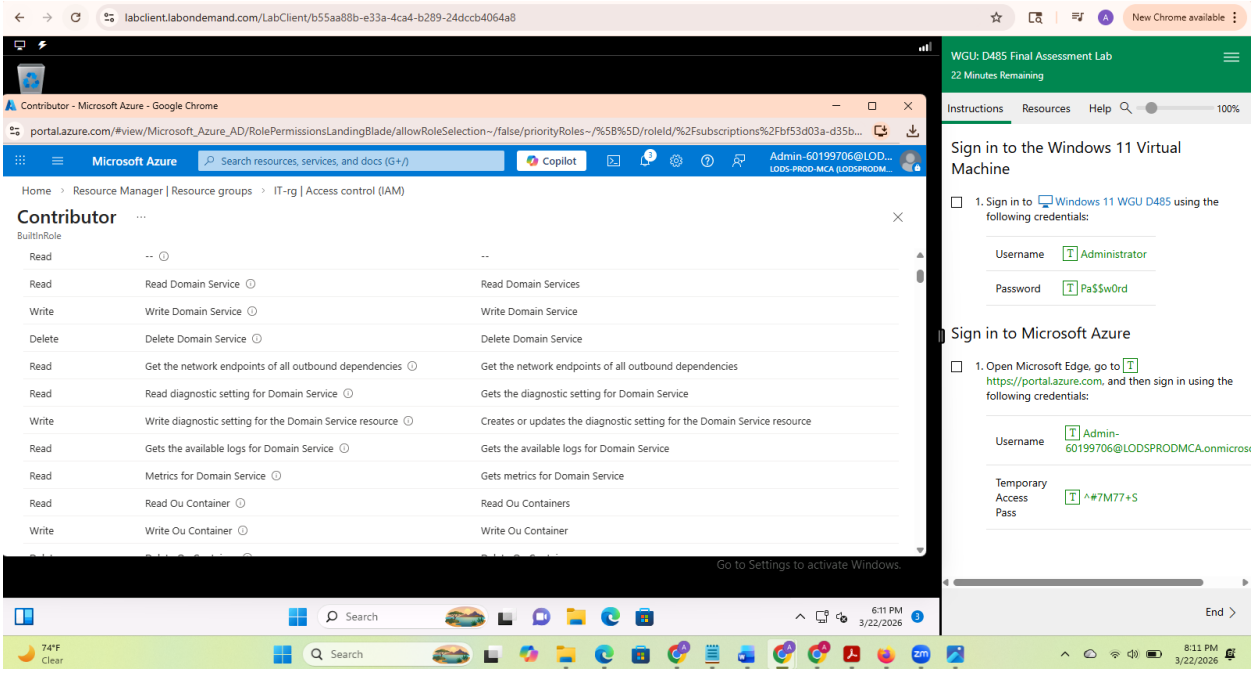
Accounting-rg RBAC:



Marketing-rg RBAC:

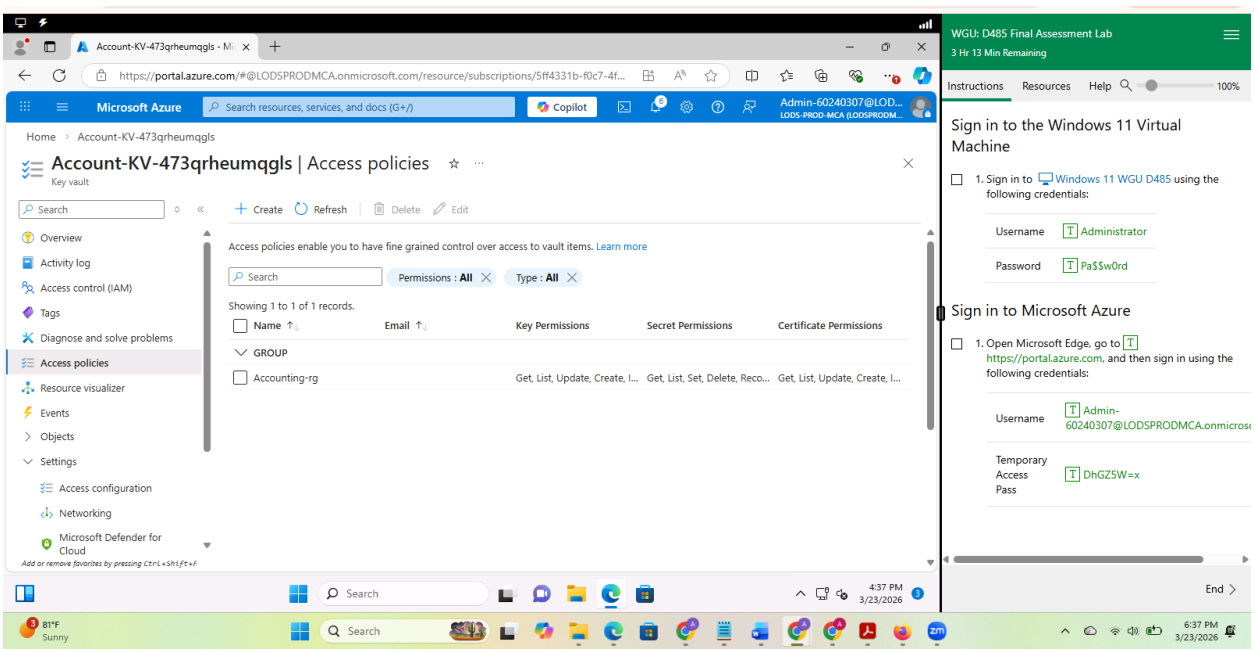


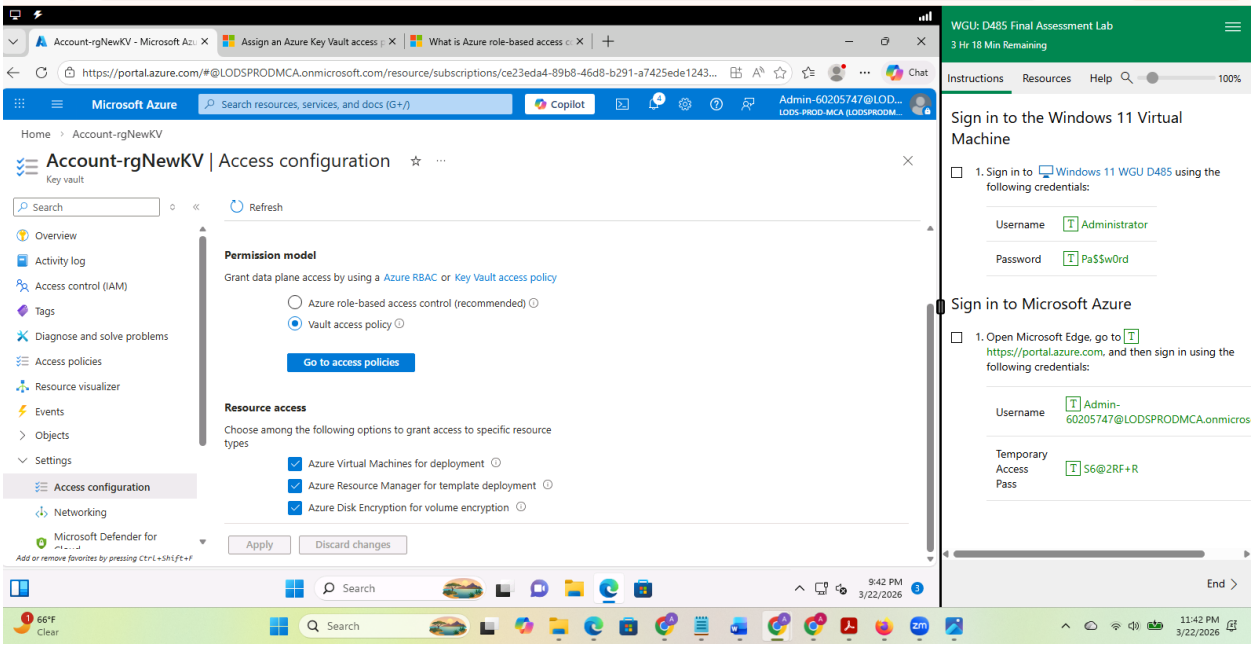
IT-rg RBAC:



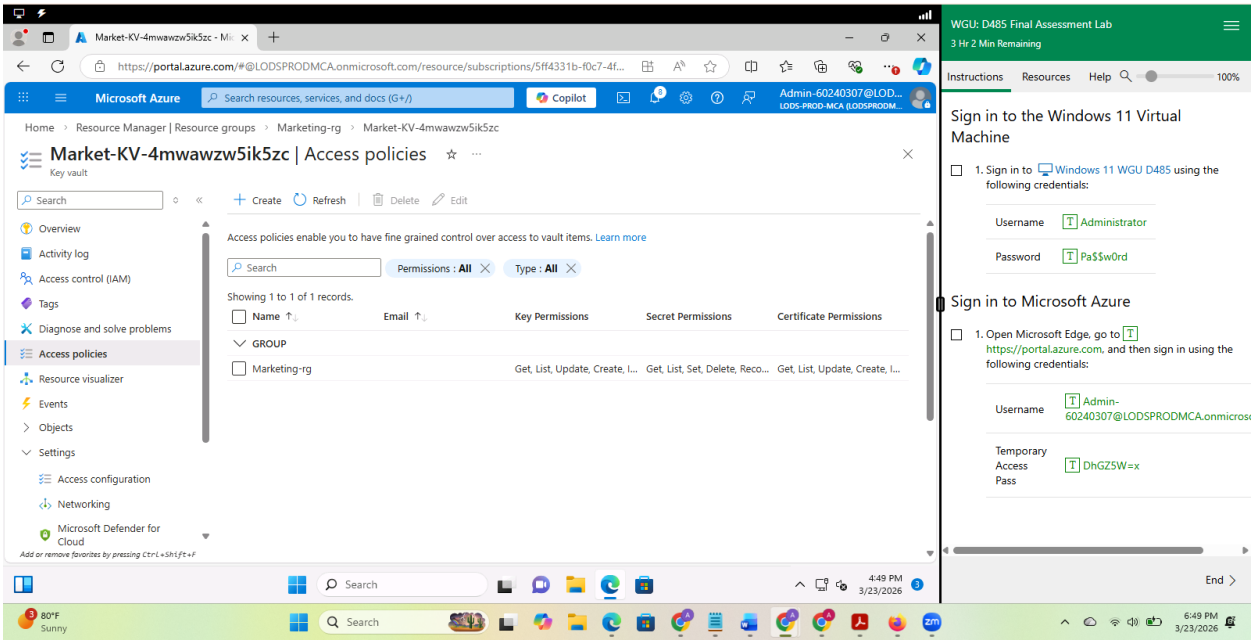
Task D:

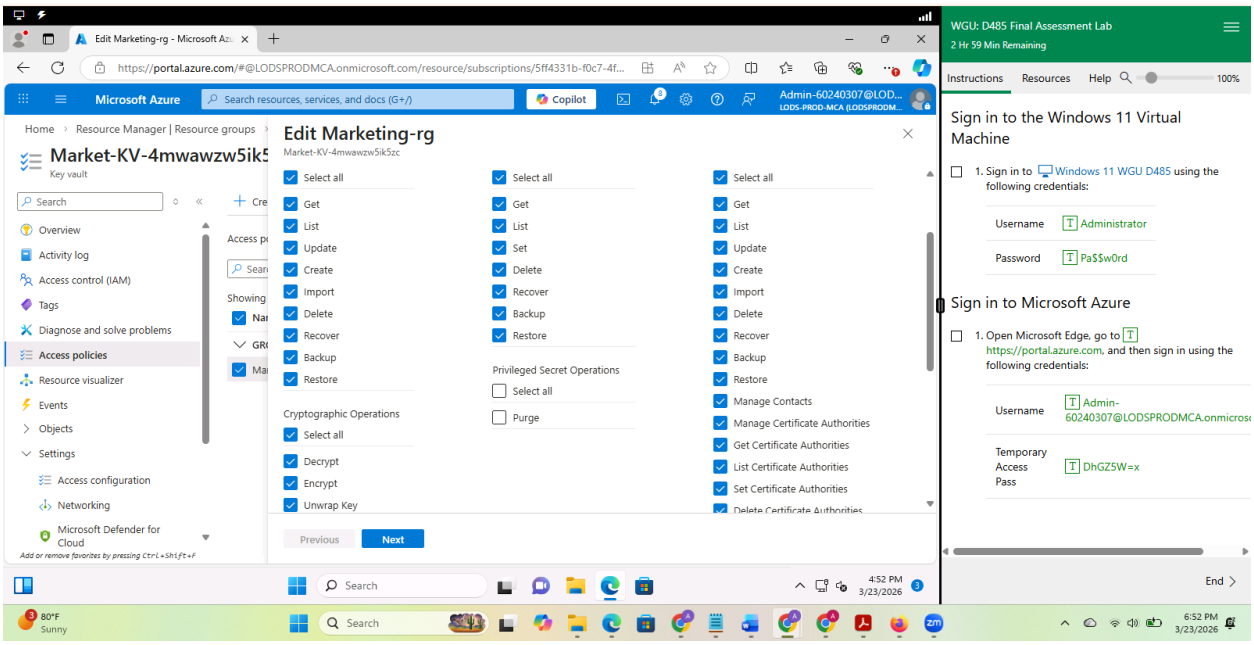
Accounting-rg Key Vaults:



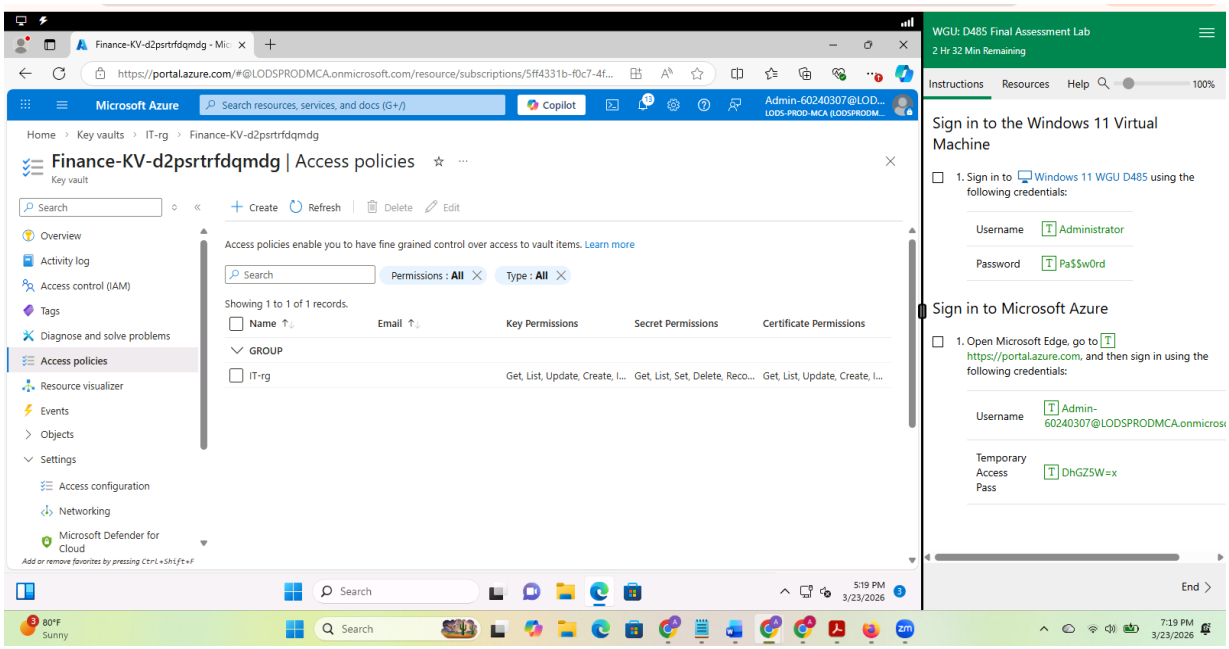


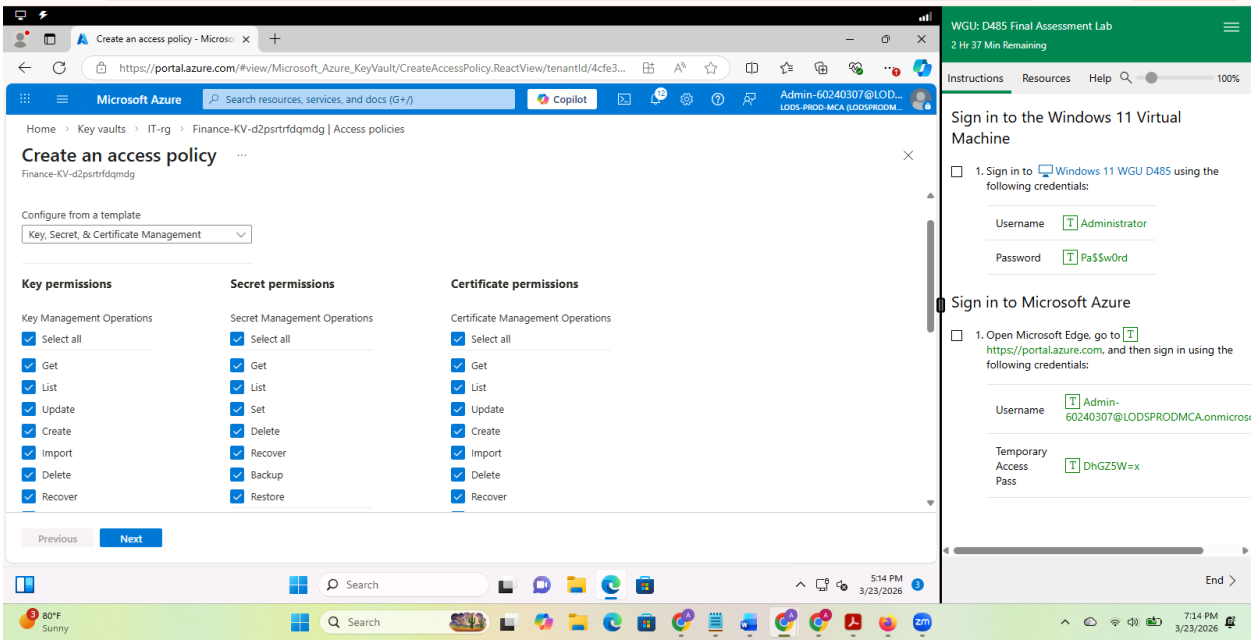
Marketing-rg Key Vaults





IT-rg Key Vaults





Task D-II

The Key Vault was misplaced between the Marketing-rg and Accounting-rg teams, which could pose a serious security risk and compliance issue. After confirming the issue, I moved both Key vaults to their appropriate resource groups to align with team workspace expectations and to enhance the security posture.

After analyzing Accounting-rg, Marketing-rg, and IT-rg, I observed that the data are not properly encrypted; as a result, I recommend and implement the following data encryption best practice.

Data at rest: Encrypting data at rest is very crucial because it helps prevent unauthorized access and use of the data. Even if an intruder manages to break into the system, we can rest assured that we have another layer of security in place while we mitigate the incident situation.

To implement this concept, using Azure Key Vault helps provide us with HSM encryption capability. Because the SWBTL LLC has to follow FISMA law, and this law requires FIP-140-2,

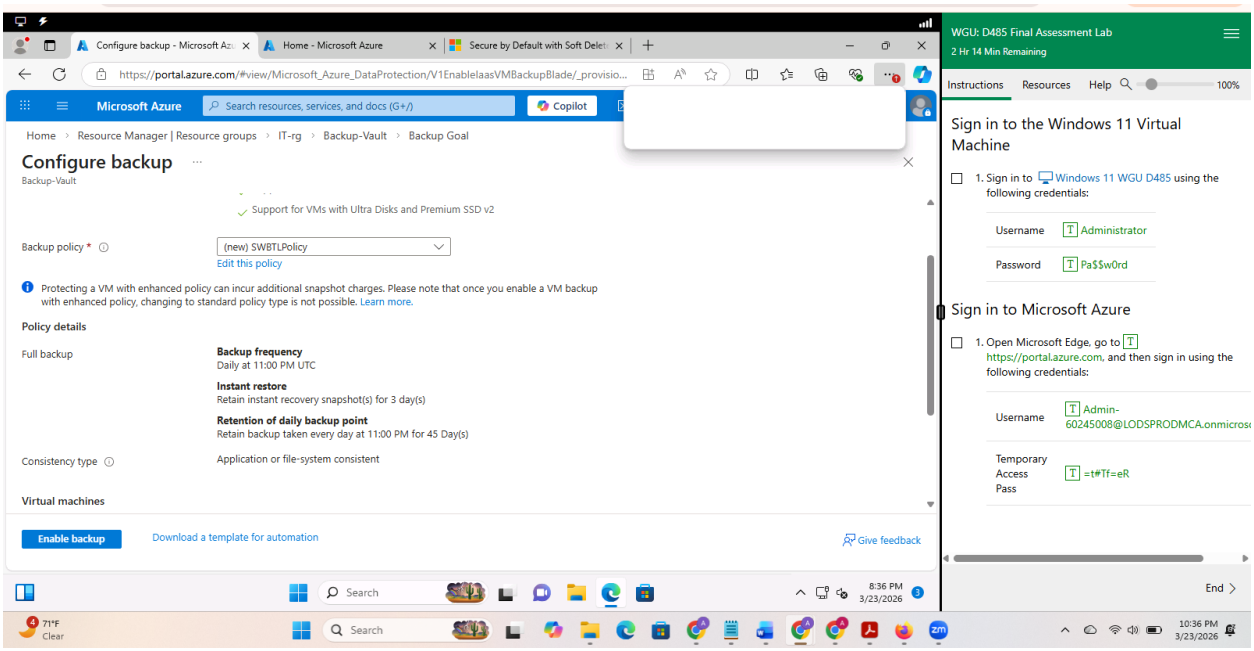
then using HSM for our data at rest is ideal and required to meet this regulation requirement and to protect our data.

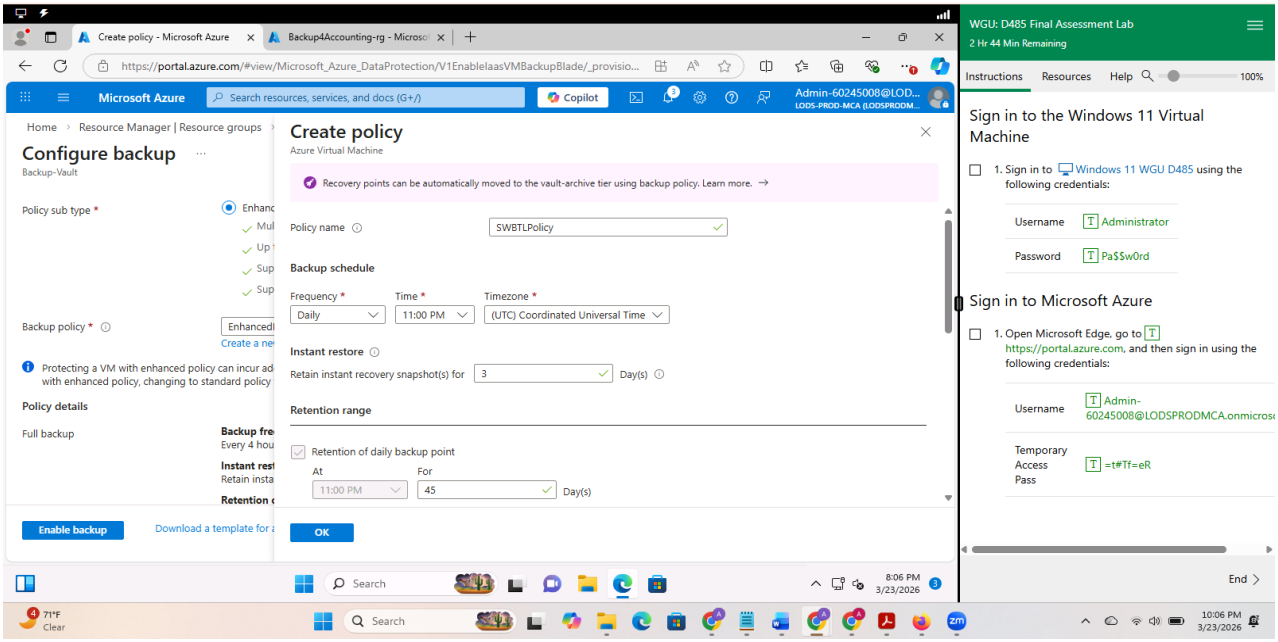
Data in transit: another data encryption best practice that helps protect our data when transferring from one location to another (Source to Destination). To achieve this objective for SWBTL LLC, I recommend implementing Azure Key Vault. With the implementation of Azure Key Vault, we can generate an SSL/TLS certificate (X.509 cert), which helps encrypt our data in transit. Not only does this satisfy the NIST 800-53 guidelines (which are referenced as a requirement for FISMA), it also satisfies the PCI DSS requirements for payment data encryption. By using Azure Key Vault, we are not only able to generate or import SSL/TLS VS 1.2 or higher, but we are also able to auto-renew and rotate the SSL/TLS certificate in use.

Task E

The backup vault was configured to satisfy the business requirement, with daily backups scheduled to run at 7 PM EST (11 PM UTC coordinated time). The retained instant recovery snapshot was configured for 3 days. Lastly, the daily backup point was configured to be retained for 45 days. All these were configured based on SWBTL LLC's business requirements. These backups are very important, especially during disaster recovery. Should the business encounter an outage or operational incident that could lead to data loss, having backups in place could help address data availability (CIA triad) for the company's customers (end users).

IT-rg for All Backups:




Create policy - Microsoft Azure
Backup4Accounting-rg - Microso
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Search resources, services, and docs (G+/)
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Home > Resource Manager | Resource groups
Configure backup
Backup-Vault
Policy sub type *
Backup policy *
Protecting a VM with enhanced policy can incur ad
with enhanced policy, changing to standard policy
Policy details
Full backup
Enable backup
Download a template for
Create policy
Azure Virtual Machine
Recovery points can be automatically moved to the vault-archive tier using backup policy. Learn more.
Policy name
SWBTLPolicy
Backup schedule
Frequency *
Daily
Time *
11:00 PM
Timezone *
(UTC) Coordinated Universal Time
Instant restore
Retain instant recovery snapshot(s) for
3
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WGU: D485 Final Assessment Lab
2 Hr 44 Min Remaining
Instructions
Resources
Help
100%
Sign in to the Windows 11 Virtual Machine
1. Sign in to Windows 11 WGU D485 using the following credentials:
Username
Administrator
Password
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Sign in to Microsoft Azure
1. Open Microsoft Edge, go to https://portal.azure.com, and then sign in using the following credentials:
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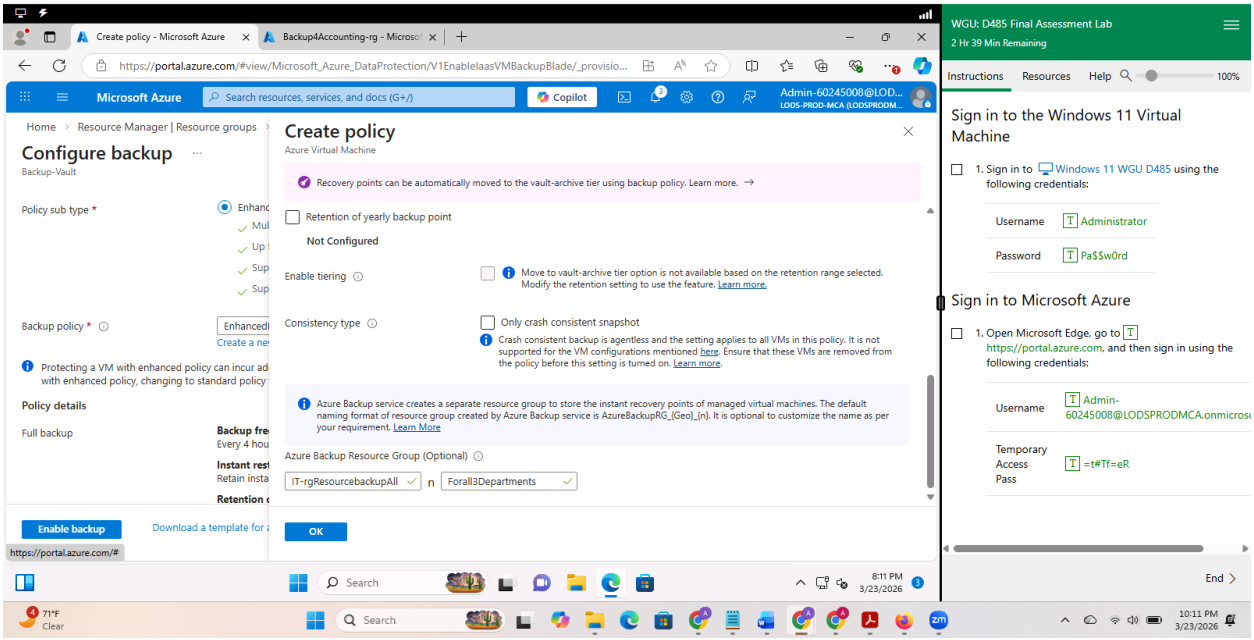
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Home > Resource Manager | Resource groups
Configure backup
Backup-Vault
Policy sub type *
Backup policy *
Protecting a VM with enhanced policy can incur ad
with enhanced policy, changing to standard policy
Policy details
Full backup
Enable backup
Download a template for
https://portal.azure.com/#
Create policy
Azure Virtual Machine
Recovery points can be automatically moved to the vault-archive tier using backup policy. Learn more.
Retention of yearly backup point
Not Configured
Enable tiering
Move to vault-archive tier option is not available based on the retention range selected. Modify the retention setting to use the feature. Learn more.
Consistency type
Only crash consistent snapshot
Crash consistent backup is agentless and the setting applies to all VMs in this policy. It is not supported for the VM configurations mentioned here. Ensure that these VMs are removed from the policy before this setting is turned on. Learn more.
Azure Backup service creates a separate resource group to store the instant recovery points of managed virtual machines. The default naming format of resource group created by Azure Backup service is AzureBackupRG_{Geo}_{n}. It is optional to customize the name as per your requirement. Learn More
Azure Backup Resource Group (Optional)
IT-rgResourcebackupAll
n
Forall3Departments
OK
WGU: D485 Final Assessment Lab
2 Hr 39 Min Remaining
Instructions
Resources
Help
100%
Sign in to the Windows 11 Virtual Machine
1. Sign in to Windows 11 WGU D485 using the following credentials:
Username
Administrator
Password
Pa$$w0rd
Sign in to Microsoft Azure
1. Open Microsoft Edge, go to https://portal.azure.com, and then sign in using the following credentials:
Username
Admin-60245008@LODSPRODMCA.onmicrosc
Temporary Access Pass
=t#Tf=eR
End

Task F/F-1

The shared responsibilities between SWBTL LLC and the cloud provider (Azure Cloud Provider) are that SWBTL LLC is responsible for the security in the cloud, while the cloud provider is responsible for the security of the cloud (The physical Infrastructure, like data center, Physical host, etc), (Share responsibility model, Microsoft 2026).

Because our cloud service model for SWBTL LLC is Infrastructure as a Service (IAAS), and we need to manage the Operating System (OS), our environment may be at risk of malware, which is a major security risk that could corrupt our servers and disrupt our deployment environments from functioning.

Another risk we may encounter when choosing the IAAS cloud service model is the need to patch for vulnerabilities. Many software and package releases issue patch updates to their applications from time to time when they discover a security vulnerability or a utility deprecation. We would need to implement these updates on our application servers, and this could become tedious if we have hundreds or thousands of servers to patch or update regularly.

The third possible risk that we may encounter is a public internet attack. Because we manage our own internet infrastructure in the cloud, we are responsible for its security in the cloud. The attackers constantly scan public cloud IP ranges for open management ports and attempt "easy" attacks, such as using common passwords and known unpatched vulnerabilities, as well as DDoS attacks, posing a potential risk to us at SWBTL LLC. (Security best practices for IaaS workloads in Azure, Microsoft 2026).

Task F – II

Mitigating the three risks identified above. First, by installing Antimalware in our system, this can help to combat the malware risk when using the IAAS cloud service model. Using Azure products such as Microsoft Antimalware for Azure, a single-agent solution for applications and tenant environments, designed to run in the background without human intervention. This is scalable and would help us to mitigate the risk.

The second risk talked about was vulnerability risk due to the need for updates or time-to-time patching based on vendor releases. We can solve this issue by enabling automatic guest patching for your Azure Virtual Machines (VMs) and Scale Sets (VMSS), which helps ease update management by safely and automatically patching VMs to maintain security compliance while limiting the blast radius. (Automatic Guest Patching for Azure Virtual Machines and Scale Sets, Microsoft 2026)

To mitigate the third risk, prevent exposure to network routing and security groups. When configuring Security groups (SGs), we need to prevent VMs from being exposed to "any" source IP address to help limit the potential attack surface. We can also use Just-in-time (JIT) to lock down inbound traffic to our Azure VMs. This approach would help reduce exposure to attacks

while providing easy access to connect to VMs when needed. (Security best practices for IaaS workloads in Azure, Microsoft 2026, https://learn.microsoft.com/en-us/azure/security/fundamentals/iaas).

Task G

With our updated cloud solution, there are some threats that we must not overlook, such as API Vulnerabilities. In the cloud, we use a microservices deployment approach (not monolithic), and all these microservices interact with each other via API calls. Exposed API endpoints could become primary targets for malicious actors or attackers.

To mitigate this threat, we can implement a countermeasure by deploying a Web Application Firewall (WAF). Using services such as Azure Front Door in front of our API Management instance to protect against common web application exploits and bots would help prevent this threat.

The second threat observed is Identity and Access Management (IAM) Risks. The attackers usually target weak credentials or misconfigured permissions in Entra ID. If some of the passwords are weak, this could pose a threat to our architectural environment. The attacker could take advantage of the opportunity to penetrate the deployment environment and become a major security issue for business operations.

To mitigate this concern, we would implement Multi-Factor Authentication (MFA) to enforce strong access controls for all logins, thereby strengthening the security posture.

The third observed threat possibility is financial threat. Over-provisioning or failing to utilize cost-management tools can result in significantly higher expenses than on-premises solutions. If the business incurs higher costs than budgeted, it could lead to an interruption in business operations, as the cloud provider may cease providing the underlying resources needed for daily operations until the bill is paid.

To prevent this threat from happening, we can use Azure Advisor to identify unused, idle, or underutilized resources such as VMs and get specific, actionable recommendations for right-sizing (Scalability) or decommissioning the excess resources. Using Azure built-in autoscaling services, such as Azure Monitor, can help in scaling in and scaling out of resources. (Overview of autoscale in Azure, Microsoft 2026, https://learn.microsoft.com/en-us/azure/azure-monitor/autoscale/autoscale-overview).

References:

[1] (2026, February 25). *Share responsibility model*. Microsoft. Retrieved March 22, 2026, from https://learn.microsoft.com/en-us/azure/security/fundamentals/shared-responsibility

[2] Microsoft (n.d.). *Security best practices for IaaS workloads in Azure*. Azure. Retrieved March 22, 2026, from https://learn.microsoft.com/en-us/azure/security/fundamentals/iaas

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[3] Microsoft (n.d.). *Automatic Guest Patching for Azure Virtual Machines and Scale Sets*. Azure. Retrieved March 23, 2026, from https://learn.microsoft.com/en-us/azure/virtual-machines/automatic-vm-guest-patching?bc=%2fazure%2fsecurity%2fbreadcrumb%2ftoc.json&toc=%2fazure%2fsecurity%2ffundamentals%2ftoc.json

[4] Microsoft (n.d.). *Security best practices for IaaS workloads in Azure*. Azure. Retrieved March 22, 2026, from https://learn.microsoft.com/en-us/azure/security/fundamentals/iaas

[5] Microsoft (n.d.). *Overview of autoscale in Azure*. Azure. Retrieved March 23, 2026, from https://learn.microsoft.com/en-us/azure/azure-monitor/autoscale/autoscale-overview